Literacy Association of Ireland Annual Conference 2026 · AI and Literacy · Saturday 7 November

Check the Machine

A classroom protocol for early critical AI literacy

Andrea C. Borowczak, Ed.D.
Professor and Director, School of Teacher Education

Mike Borowczak, Ph.D.
Associate Professor of Electrical and Computer Engineering

University of Central Florida

Before step: write your own answer first



Check the Machine

The task

Explain to a 5th class pupil why we have seasons.

- Write one sentence on your own: 30 seconds
- Compare with the two people beside you
- Then we show what one AI wrote

After step: what one AI wrote

Task → Before → **After** → Takeaway

Check the Machine

> “Earth experiences seasons because of its elliptical orbit around the Sun. When Earth is closer to the Sun, we have summer, and when it is farther away, we have winter. That’s why the Northern Hemisphere is warmer in July and colder in January — Earth’s distance to the Sun changes throughout the year.”

AI-generated explanation, from the AI Error Gallery of the CRAFT PD Series (cxedhub.com)

- Where does it match your sentence?
- Which sentence would you check, and against what?

Takeaway step: what writing first let you see

Task → Before → After → **Takeaway**

Check the Machine

Early January

Earth's closest approach to the Sun, in the middle of our winter

Late June

Limerick's longest days and Sydney's shortest, with one Sun and one Earth

Seasons come from the tilt of Earth's axis. A committed answer gave every reader something to check the paragraph against.

Check the Machine: four steps

1

Task

State what you asked the tool to do

→

2

Before

Commit to your own answer before the tool answers

→

3

After

Compare the tool's output with your answer

→

4

Takeaway

Record what the gap shows about the tool, and about you

AI is never the first step

Errors are features

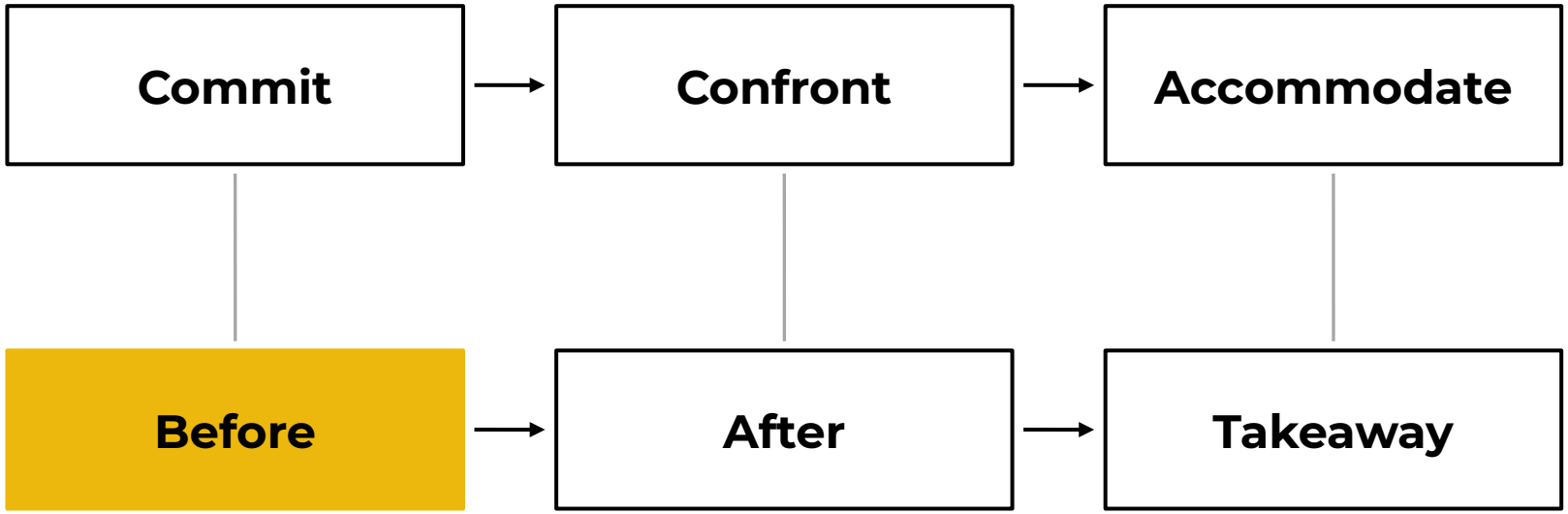
The check travels to the next tool

C R A **F** T

The routine inside the Fortify phase, where the CRAFT cycle checks the work

Why the Before step comes first

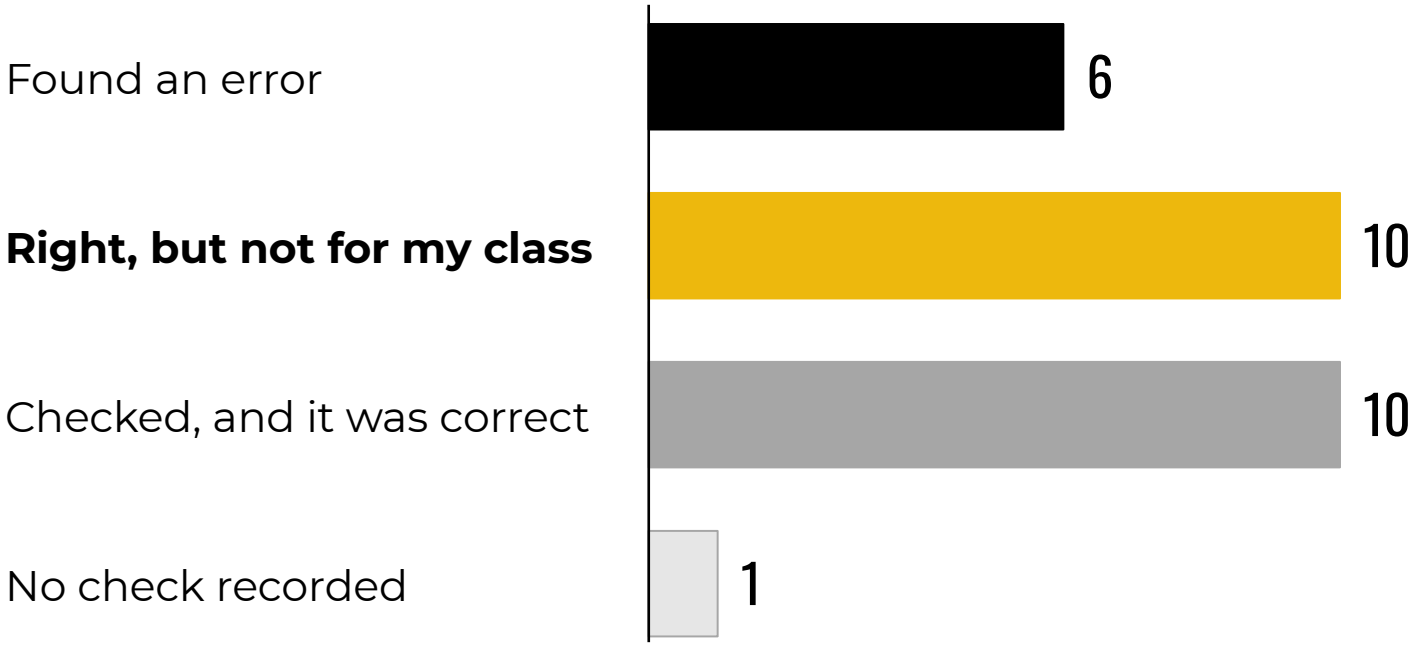
Conceptual change



Check the Machine

- A tutor that required reasoning before answers removed the exam penalty (Bastani et al., 2025)
- Checking starts as a social act: groups of three or more first, then alone
- The same move as the Reframe phase, aimed at a tool's output

Teachers tried to break an AI in their own subject



What they caught

- A square root off by more than 3
- A geometry proof whose first line was false
- Energy flow and nutrient cycling merged into one process
- Citations that did not hold up

√2,420,520: the AI gave 1,559.01; a calculator gives 1,555.80

Workshop 2, 18 April 2026 · 27 attempts by K–12 teachers, each with a chatbot of their choice · first-pass coding by the presenters

Right, but not for my class

> “The answer is correct but over-detailed. None of my students would give me a detailed answer like what I got from ChatGPT.”

High school mathematics teacher

> “It used different symbols than those I use in my math classroom.”

High school mathematics teacher

> “We did not include ‘make middle school friendly’ so information provided veered into high school level content…”

Middle school science teachers

> “Bias: Canon White-Male European (honorary Madame Curie inclusion)”

Science teachers, on an AI-written history-of-science activity

The check covers fit to the learner along with truth: level, notation, method, and whose voices appear

Workshop 2 shared group workspace, 18 April 2026 · quoted with light punctuation edits, names withheld

First runs: most Before-step entries described a format



4 committed answers

12 expected a format or coverage

No run recorded a factual error; Takeaway-step notes were about the prompt, the level, or the lesson.

> “Mass does not change – this is a physical change”

A committed answer

> “A number that is in hertz or HZ”

An expected format

> “Expectations are visuals and maybe video.”

An expected format

For children the Before step is an answer: said, drawn, or written first

Workshop 2 partner practice, 16 logged runs with a Before-step entry · quoted with light punctuation edits · first-pass coding by the presenters

One teacher's version for students

Task	What specific science question or command did you give the AI? Paste your prompt here so you remember exactly what you asked about EM waves.
Before	Think about what you already know about the EM spectrum. Based on your science notes and brainpower, what facts do you expect to see?
After	Read the answer carefully. Does it match your science knowledge? Highlight any parts that seem off…
Takeaway	Did the AI hallucinate a science fact or get a calculation wrong? Did this help you remember a concept better…?

What the teacher added

- A source for the Before step: your notes
- An action for the After step: highlight
- Two questions for the Takeaway step: about the tool, and about you
- Words students can use: “seem off”, “hallucinate”

Written during Workshop 2, 18 April 2026, by a participating science teacher · quoted as written, trimmed

What the early-grades teachers did

Elementary group	Planned a kindergarten check of “heavier objects fall faster”: drop two objects Design the check too: use two balls; in air a feather falls slowly
Kindergarten	Asked for a problem that does not fit two standards, then judged the AI's reasoning
Grade 2	Asked for an array of 8 red and 8 blue cards; noticed it showed only one possible array
Grade 3	Solved the AI's fence-length word problem by hand and checked the teacher's edition
Elementary group	Norm: “Filter information: use a rubric and cite source.”

Workshop 2, 18 April 2026 · four of the 29 teachers taught kindergarten to grade 3; the elementary group worked together · names withheld

Designing it for young children

- Same four steps every time, short enough to repeat daily
- The teacher holds the screen; children meet the output read aloud or on a printed card
- The Before step is said, drawn, or written before anything is shown
- Check against something concrete: an object, a book, a measurement
- Groups of three or more first, then alone

A talk frame for the Takeaway step

The machine said ___.
I think ___
because ___.

Our proposal for children to voice disagreement in their own words

Templates openly licensed (CC BY-SA 4.0) at cxedhub.com, written to be adapted across grades and languages

Next in this session: GenAI literacy in teacher education, Ó Ceallaigh and Murphy

Check the Machine

Teach learners to trust AI output as far as the evidence supports.

Andrea C. Borowczak · Andrea.Borowczak@ucf.edu
Mike Borowczak · mike.borowczak@ucf.edu

Check the Machine template and workshop materials: cxedhub.com